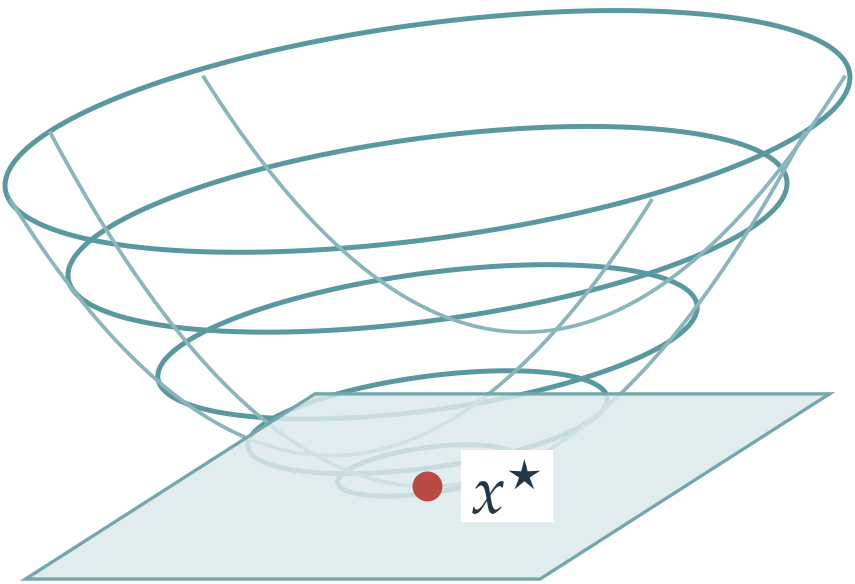
Convex objective



$$\nabla f(x^\star) = 0$$

A global minimizer